

Crossdem: Comparing Politicians' Rhetoric With Democracy Indicators

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Comparison between politicians' speeches and V-Dem democracy indices. After building a corpus of public speeches from Italian Prime Minister, U.S. Presidents and French Prime Ministers and Presidents, NLP analyses will be performed, from text complexity to hate speech recognition. The ensuing results will be compared with V-Dem democracy indices to shed light on possible correlations. Lastly, a transformer model will be fine-tuned to interpret political language and classify hate speech.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; • **Social and professional topics** → **Hate speech**; **Political speech**.

Additional Key Words and Phrases: AI, NLP, Politics, Speeches

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1 INTRODUCTION

Politicians are some of the most influential people in human society: from the Greeks' demagogues to our Presidents and Prime Ministers, leaders have always tried to steer society. Shapiro and Page (1984) [29] showed how nearly all 20th century U.S. Presidents have tried to lead public opinion, with varying degrees of success. As societies became more democratic, so grew the leaders' responsibility to convince potential followers that their policies are beneficial. Thus, speeches play a vital role in the functioning of society [3]: leaders depend on the verbal power to persuade people [8].

Moreover, democratic stability depends on citizens on the losing side accepting election outcomes. Clayton et al. [10] evaluated the effects of exposure to multiple statements from (at the time) former president Donald Trump attacking the legitimacy of the 2020 US presidential election. They found no evidence indicating that elites can erode democratic norms easily or that the effects of norm violations are uniform across the entire population. However, elite rhetoric can shape normative beliefs in core democratic values such as confidence in elections and support for peaceful transfers of power. This was especially evident amongst people who approved of Trump's performance in office.

Having established the importance of speeches, we can differentiate between two types: *internal speeches*, aimed at other institutional bodies, such as parliamentary speeches, and *external speeches*, aimed at the population, such as public speeches, interviews and social network posts. This last category is especially interesting since it affords politicians a direct and high-volume communication channel to their potential followers: U.S. President Donald Trump tweeted 57,000 times between May 2009 and January 2021 alone [25], while Italy's Prime Minister Giorgia Meloni totalled more than eighty million social network interactions in 2025 [36].

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On the other hand, internal speeches are especially easy to access, being already aggregated in public databases, yet their technical and codified nature makes their interpretation harder for both human and automated agents. Moreover, until 1985 Italian stenographic documentation did not report speeches word for word, but rather translated them into a "language of the parliament" [19, 27].

2 SOCIAL NETWORKS

Broadly speaking, the goal of this mini-project is to gather a corpus of politicians' speeches and perform linguistic analyses on it. The idea is to start from standard NLP tasks such as word frequency and text complexity, to then move to classification tasks such as sentiment analysis and hate speech detection.

This objective is why I ultimately decided to gather a corpus of *external speeches* rather than internal ones. Large language models are trained with tokenizers, and the resulting token distribution is highly imbalanced: Chung et al. [9] did a controlled study that scaled the vocabulary of the language model from 24K to 196K while holding data, computation, and optimization unchanged. They discovered that models are disproportionately optimized on the part of language that appears most often, meaning they are way better at understanding common words and patterns. This translates to the fact that internal speeches are less understandable for an AI agent.

As previously stated, external speeches include social network communications. While television is still the first source of information for the majority of the population, online platforms are steadily growing and are already the preferred media for young people [15, 26]. There is evidence of an increase in political participation due to social media usage, as well as risks for the functioning of democracy [2, 24], and there have been attempts to predict election results with social media data analyses. Rita et al. measure sentiment polarity on Twitter, and conclude that tweets' sentiment is not a reliable election results predictor. Additionally, results also show that it is impossible to state that social media impacts voting decisions [33].

Opposite results can be seen in Belcastro et al.'s [5] 2022 study: a real-time analysis was carried out during the 2020 US presidential election campaign, correctly identifying the leading candidate before Election Day in 10 out of 11 swing states.

Silva et al. [35] managed to match tweets against parliamentary speeches to measure politicians' sentiment on a specific topic. They find that parliament members who participate less in parliamentary debate tend to have larger differences with their party on Twitter, suggesting that a certain level of self-censoring is taking place in the parliamentary arena.

That being said, X just one channel among many used to connect with the voters: official messages to the nation, interviews, and public speeches are still powerful tools, and are often made publicly available in video format by the politicians themselves on their Facebook page and Youtube channel.

3 PROJECT GOAL

Let's now define the goal of the mini-project.

I want to compare the rhetoric between the first and the current¹ Italian Prime Ministers, namely Alcide De Gasperi and Giorgia Meloni. I also want to produce a proof-of-concept for a speech-to-text pipeline that, given a set of URLs, each of which a Youtube's permalink, transcribe the video into text.

¹At the moment of writing: May 2026.

The rhetoric will be compared via NLP tools and pipelines, such as measuring textual complexity and classifying populism or hate speech.

3.1 Roadmap

The first step is building a corpus of politicians' speeches, one corpus for Italian speeches, one for French speeches and one for American speeches. To limit the scope, I decided to take only speeches from Italian Prime Ministers, U.S. Presidents, and either French Presidents when the President and the Prime Minister come from the same party, or French Prime Ministers when the President and Prime Minister are in opposition to each other. This is justified by the fact that when they are politically aligned, the President dominates. During cohabitation (opposing political camps), the Prime Minister gains real policy autonomy [28].

Speeches can be taken from different sources, such as official speeches, interviews, or social network posts, and can be in different media formats, such as textual, audio or video. For the latter two, a translating tool will be needed (such as a speech-to-text library).

The second step is using the corpus to perform linguistic analyses. The corpus can be subjected to data augmentation², since we can expect the density of speeches to decrease as we go back in time. Then, the corpus will be pre-processed, with procedures such as tokenization, stemming and lemmatization. We can then compute word frequency and text complexity measures, as well as hate speech and populism identification.

The third step is cross-referencing the linguistic metrics with democratic indicators such as freedom of expression, freedom of association, transparency and fairness of law and other such metrics. All democracy metrics can be found in the V-Dem dataset.

The fourth step is to fine-tune a pre-trained transformer model, for example Mistral 7B³, to make it better understand political speeches in three different languages, and to train it in a classification task for hate speech and populism recognition.

4 BUILDING THE CORPUS

The natural first step was to look at other corpora that aggregate similar data.

The **Europarl** dataset [22] aggregates proceedings of the European Parliament in twenty-one different European languages⁴ with the goal of generating sentence-aligned text for statistical machine learning translation systems. The overall structure is akin to a collection of plaintext with minimal tagging.

ItaParlCorpus [12], on the other hand, is a comprehensive, annotated, and machine-readable database of Italy's parliamentary speeches spanning from 1948 to 2022. Each speech is identified by a precise date, the role of the speaker, their party name and family, and the legislature number. It is not, however, very rich in terms of linguistic processing tagging.

IMPAQTS [11] is a multimodal corpus of around 2.65 million tokens, including 1,500 speeches uttered by 150 prominent politicians, spanning from 1946 to 2023. For each speaker, the corpus contains 4 parliamentary speeches, 2 rallies, 1 party assembly, and 3 statements (in person or broadcasted). The goal is to annotate political rhetoric that hides a deeper meaning, for example by employing sarcasm. There are four main types of annotation: implication

²In its most general sense, data augmentation denotes methods for supplementing so-called incomplete datasets by providing missing data points in order to increase the dataset's analyzability [21].

³Mistral 7B the most powerful language model for its size to date: <https://mistral.ai/news/announcing-mistral-7b>.

⁴All 21 Europarl languages: Romanic (French, Italian, Spanish, Portuguese, Romanian), Germanic (English, Dutch, German, Danish, Swedish), Slavic (Bulgarian, Czech, Polish, Slovak, Slovene), Finno-Ugric (Finnish, Hungarian, Estonian), Baltic (Latvian, Lithuanian), and Greek.

(conventional, generalized, etc.), presupposition (e.g., pragmatic), vagueness (syntactic, semantic, metaphoric), and topicalization (either syntactic or prosodic).

Parola di Leader, and specifically the corpus LP4 [16], is a corpus of more than five million occurrences taken from the stenographic archives of the Chamber of Deputies of Italy⁵, between 1948 (first legislature of the Italian Republic) and 2011 (excluding the Monti Government, the 61st legislature).

They selected a list of thirty "highly influential politicians"⁶ and then for each of them they selected some speeches. Overall, they collected 1877 distinct speeches.

The corpus could be considered a collection of thirty distinct corpora, one for each politician. Each of these has, among others, the following features: name of the politician, role of the politician (for example, Prime Minister), whether the politician is in the majority party or in the opposition, legislature number and sitting number of the day, date of the sitting, topic of the speech, name of the current Government (e.g., *Berlusconi I*), and lastly a reference to the file that holds the actual speech.

There is also a corpus for the vocabulary that holds, for each word, the total number of occurrences, a part-of-speech⁷ tag, its lemma, a count for the number of occurrences for each politician, majority party and opposition, and lastly the imprinting tag (e.g., masculine singular).

Lastly, **ParlaMint** [13, 14] is a compiled collection of comparable parliamentary corpora for a number of countries and languages. ParlaMint corpora are interoperable, i.e. encoded to a very constrained common ParlaMint schema, a specialization of the Parla-CLARIN⁸ recommendations, which are a customization of the TEI Guidelines⁹.

It is a collection of parliamentary sittings split into utterances. An example of two utterances could be *Politician: "I ask permission to speak."*, *President of the Senate: "Permission granted."*. For each utterance there are, among others, the following metadata: an ID for both the whole sitting and the utterance itself, a title of the sitting, a date of the sitting, the State Body where it took place (e.g., the Senate of the Republic), the name of the speaker and their affiliation, as well as the name of their coalition.

Speeches are collected before being split into their own datasets, and for each speech we have, among others, a unique ID and a date, the type of speech (e.g., *Interview* or *Rally*), the venue channel (e.g., *BBC News*), the unique ID of the speaker and their role (e.g., *host* or *main speaker*), the text of the speech itself and its type (e.g., *Question*, *Answer*, *Interruption*) and whether there have been any *Incidents* and their types.

Incidents are breaks in the flow of the conversation, and they can be of different types¹⁰, for example *"leaving"* or *"break"*. If an incident is of the kinesic type, it can be further subdivided into different kinesic types, such as *"applause"* or *"laughter"*¹¹. If the incident is of the vocal type, it can be further subdivided into different vocal types, such as *"question"* or *"shouting"*¹².

Speakers and Parties also have their own datasets, with unique IDs, names, affiliations, roles and the like.

⁵Chamber of Deputies official online public archive: <https://legislatureprecedenti.camera.it/>.

⁶The full list of politicians follows: Giorgio Almirante, Giuliano Amato, Giulio Andreotti, Enrico Berlinguer, Silvio Berlusconi, Pier Luigi Bersani, Fausto Bertinotti, Rosy Bindi, Emma Bonino, Umberto Bossi, Pier Ferdinando Casini, Francesco Cossiga, Bettino Craxi, Massimo D'Alema, Alcide De Gasperi, Ciriaco De Mita, Antonio Di Pietro, Amintore Fanfani, Gianfranco Fini, Ugo La Malfa, Aldo Moro, Pietro Nenni, Achille Occhetto, Marco Pannella, Romano Prodi, Giuseppe Saragat, Giovanni Spadolini, Palmiro Togliatti, Valter Veltroni, Nichi Vendola.

⁷In grammar, a part of speech or part-of-speech (abbreviated as POS or PoS) is a category of words (or, more generally, of lexical items) that have similar grammatical properties [30].

⁸Parla-CLARIN, a TEI schema for corpora of parliamentary proceedings: <https://clarin-eric.github.io/parla-clarin/>.

⁹P5 TEI Guidelines: <https://tei-c.org/guidelines/p5/>.

¹⁰Full list of incident types: "action", "incident", "leaving", "entering", "break", "pause", "sound", "editorial".

¹¹Full list of kinesic types: "kinesic", "applause", "ringing", "signal", "playback", "gesture", "smiling", "laughter", "snapping", "noise".

¹²Full list of vocal types: "greeting", "question", "clarification", "speaking", "interruption", "exclamat", "laughter", "shouting", "murmuring", "noise", "signal".

Overall, the taxonomy of ParlaMint is strongly codified, which makes the dataset machine-readable and easier to interpret. All data is also saved in XML format.

4.1 My Corpus

Ultimately, I did not manage to find any corpus that I could directly integrate or use for my purposes, but I took inspiration from ParlaMint for the overall structure. At the time of writing, the corpus is not yet compiled, so the structure may change. The structure for the Italian corpus is the only one present at the moment. The structure is explained as tables and columns (or *features*) of said tables.

Italian Government Table:

- Column *government*:
 - e.g., `degasperi2`: "De Gasperi II";
 - e.g., `spadolini1`: "Spadolini I";
 - e.g., `forlani`: "Forlani";
- Column *primeMinister*:
 - e.g., `degasperi:prime_ministers.id==degasperi`;
- Column *dateStart*: ISO 8601 (YYYY-MM-DD);
- Column *dateEnd*: ISO 8601 (YYYY-MM-DD);
- Column *legislature*:
 - e.g., 1: "I Legislatura";
 - e.g., 15: "XV Legislatura";
- Column *pm_party*: The political party of the Prime Minister;
- Column *gov_leaning*: The political leaning of the specific government.

Prime Ministers Table:

- Column *pm_ID*:
 - e.g., `degasperi`;
 - e.g., `draghi`;
 - e.g., `meloni`;
- Column *pm_name*:
 - e.g., Alcide;
 - e.g., Mario;
 - e.g., Giorgia;
- Column *pm_familyName*:
 - e.g., De Gasperi;
 - e.g., Draghi;
 - e.g., Meloni;
- Column *pm_birthDate*: The birth date of the Prime Minister;
- Column *pm_deathDate*: The death date of the Prime Minister.

The dataset will have a structured taxonomy; for example, the political leaning will be codified as follows:

- mixed: e.g., CLN¹³, Pentapartito¹⁴;
- RR: Far-Right;
- LL: Far-Left;
- right: Right;
- left: Left;
- center: Center;
- centerRight: Center-Right;
- centerLeft: Center-Left.

4.2 Data Augmentation and Synthetic Data

It is likely that for older politicians there are fewer speeches available, i.e., the dataset becomes more sparse as we go back in time. Moreover, speeches will be in three languages, namely Italian, French and English. For this reason, data augmentation techniques may be useful (figure 1).

Data augmentation uses pre-existing data to create new data samples that supplement so-called incomplete datasets by providing missing data points, improving machine learning model optimization and generalization. In other words, data augmentation can reduce overfitting and improve model robustness. This is different from generating synthetic data, which is entirely artificial: data augmentation copies existing data and transforms those copies to increase the diversity and amount of data in a given set.

Rule-based approaches include relatively simple find-and-replace techniques, such as random deletion or insertion. In this strategy, one or more words in a string are replaced with their respective synonyms as recorded in a predefined thesaurus, such as WordNet or the Paraphrase Database. Back-translation (figure 1b) is a technique where the corpus is translated into another language before being translated back to the original language [21].

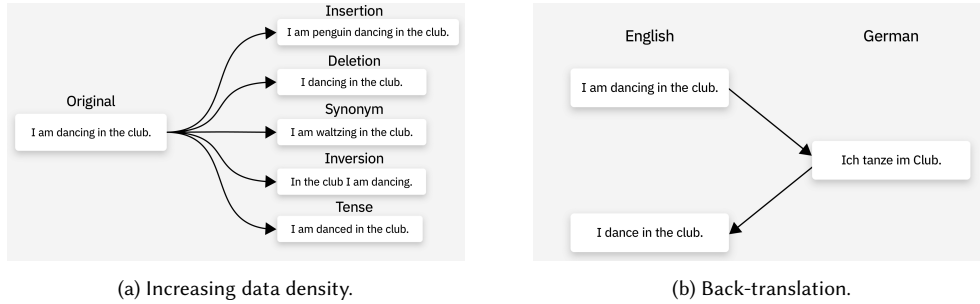


Fig. 1. Data augmentation techniques, such as synonym replacement and back-translation. Source: Kavlakoglu, IBM [21].

Synthetic data is inherently privatized, since it cannot be ascribed to any real person. This makes it free from any GDPR control system and privacy concerns. It can be of two types: fully synthetic, which does not include any real-world information, or partially synthetic, where portions of the original dataset are replaced with artificial values. Usually, it is generated with either GANs or transformer models and comes with two main problems: bias and model collapse.

¹³The National Liberation Committee (Italian: Comitato di Liberazione Nazionale, CLN) was a political umbrella organization and the main representative of the Italian resistance movement fighting against the occupying forces of Nazi Germany and the fascist collaborationist forces of the Italian Social Republic. It was composed of Christian Democrats, Socialists, Communists, and Liberals.

¹⁴The Pentapartito was the coalition government of five Italian political parties that formed between June 1981 and April 1991. It was composed of Christian Democrats, Social Democrats, Liberal Democrats, and Conservative Liberals.

Synthetic data can exhibit biases present in the real-world data that the model is based on, and when an AI model is repeatedly trained on AI-generated data, its performance declines quickly [6, 20, 32].

We probably will not need the enhanced privacy granted by synthetic data since the dataset will be compiled using public speeches from public figures.

5 LINGUISTIC ANALYSIS

Sawicki et al. 2023's survey [34] tells us that English is by far the most represented language in the field of natural language processing. Fortunately, French and Italian are still in the top eight (figure 2), so finding libraries already capable of handling them should be easy enough. It is worth noting that results could be biased since the authors looked for papers written in English.

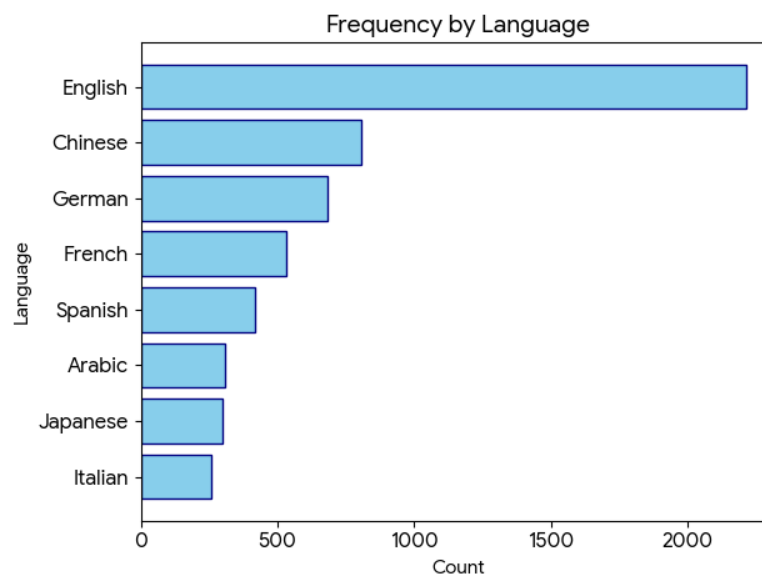


Fig. 2. Most popular languages in the field of Natural Language Processing [34]

The pipeline for pre-processing is quite standard and involves, among other steps, removing stop words, part-of-speech tagging, tokenization and lemmatization, as well as building text embeddings.

FastText¹⁵ is an open-source, free, lightweight library that allows users to learn text representations and text classifiers; it gained significant popularity due to its performance in computing text embeddings.

LLM2Vec [4] is a recipe to convert decoder-only LLMs into text encoders so that they can be used to generate state-of-the-art text embeddings.

Stanza¹⁶ is a collection of accurate and efficient tools for the linguistic analysis of many human languages. Starting from raw text, Stanza divides it into sentences and words, and can then recognize parts of speech and entities, perform syntactic analysis, and more. Stanza brings state-of-the-art NLP models to the languages of your choosing.

¹⁵FastText is a lightweight library for efficient text classification and representation learning: <https://fasttext.cc/>.

¹⁶Stanza – A Python NLP Package for Many Human Languages: <https://stanfordnlp.github.io/stanza/index.html>.

spaCy¹⁷ is another well-known and powerful NLP library for Python that supports multiple languages. It also supports tokenization, part-of-speech tagging, dependency parsing, sentence segmentation, text classification, and lemmatization. It supports multi-task learning with pretrained transformers, includes pretrained word vectors, supports custom models in PyTorch and TensorFlow, and has built-in syntax visualizers.

simplemma¹⁸ is also worth mentioning as a lightweight Python lemmatizer.

textacy¹⁹ is a Python library for performing a variety of natural language processing (NLP) tasks. It is built on spaCy and intended to be used both before (pre-processing step) and after (NLP step) it. It assesses text complexity with the Flesch-Kincaid readability test and text diversity with the Type-Token Ratio.

Text complexity can also be measured with **TRUNAJOD**²⁰, another spaCy-based Python library that implements synonym overlap, coherence between sentences, and other metrics.

Regarding populism and hate speech classification, there are many transformer-based models available from **Hugging Face**²¹. Some examples follow:

- English: <https://huggingface.co/Hate-speech-CNERG/dehatebert-mono-english>;
- French: <https://huggingface.co/Hate-speech-CNERG/dehatebert-mono-french>;
- Italian: <https://huggingface.co/Hate-speech-CNERG/dehatebert-mono-italian>.

There are transformer models based on RoBERTa for both hate speech and populism classification. For example, the following is a 0.4 billion parameter model based on Trump speeches: <https://huggingface.co/coastalcph/roberta-large-ft-trump-populism>.

Hate speech can also be detected with the Python library **HateSonar**²², which uses a BERT-based model.

5.1 Speech To Text

Since there are plenty of video and audio speeches publicly available, they could be integrated into the corpus after transcribing them into text. This can be achieved with libraries such as **SpeechRecognition**²³, which supports several different engines and APIs both online and offline, or OpenAI's **Whisper**²⁴ library. Whisper is a Transformer sequence-to-sequence model trained on various speech processing tasks, including multilingual speech recognition, speech translation, spoken language identification, and voice activity detection. It comes in six different model sizes; the "small" model requires approximately 2GB of VRAM, making it easy to run locally.

Whisper's performance varies depending on the language. Italian is the fourth best-performing language with a WER²⁵ of 5.5%. English follows at 9.3% and French at 10.8% [31]. The worst-performing language is Albanian with a WER of 55.7%.

Whisper can be used both in Python scripts and via a command-line interface.

¹⁷spaCy, Industrial-Strength Natural Language Processing: <https://spacy.io/>.

¹⁸simplemma is a lightweight toolkit for multilingual lemmatization and language detection: <https://pypi.org/project/simplemma/>.

¹⁹textacy: NLP, before and after spaCy. <https://pypi.org/project/textacy/>.

²⁰TRUNAJOD: A text complexity library for text analysis built on spaCy. <https://trunajod20.readthedocs.io/en/latest/>.

²¹Hugging Face, Inc., is an American company based in New York City that develops computational tools for building applications using machine learning. Its transformers library, built for natural language processing applications, and its platform allow users to share machine learning models and datasets and showcase their work. <https://huggingface.co/>

²²HateSonar: Hate Speech Detection. <https://pypi.org/project/hatesonar/>.

²³SpeechRecognition, a Python library for performing speech recognition with support for several engines and APIs, online and offline. <https://pypi.org/project/SpeechRecognition/>.

²⁴Whisper is a general-purpose speech recognition model. <https://github.com/openai/whisper>.

²⁵Word Error Rate (WER) is a common metric for the performance of speech recognition or machine translation systems. It is measured as an error rate percentage for compared sentences.

6 DEMOCRACY INDEX

There are plenty of sources for measuring democracy levels. Our World in Data mainly uses six: Varieties of Democracy (V-Dem), the Lexical Index of Electoral Democracy (LIED) by Skaaning et al. (2015), Freedom House's (FH) Freedom in the World index, the Bertelsmann Transformation Index (BTI) by the Bertelsmann Foundation, the Economist Intelligence Unit's (EIU) Democracy Index, and Polity by the Center for Systemic Peace [17].

Out of the six mentioned datasets, only V-Dem and LIED have data spanning between 1946 and 2025. However, V-Dem offers more granularity and is generally considered a more complete and robust dataset. It is the best choice if we are interested in both large and small differences in varieties of democracy far into the past, or if we want to use country experts to measure characteristics of political systems that are difficult to observe. These experts are anonymous and are primarily academics or members of the media and civil society. They are also often nationals or residents of the country they assess; therefore, they know its political system well and can evaluate aspects that are difficult to observe. V-Dem's own team of researchers supplements these expert evaluations [18].

6.1 Varieties of Democracy Dataset

All data is available for download at <https://v-dem.net/data/the-v-dem-dataset/> or can be accessed directly as an R package.

V-Dem distinguishes between *indicators*—lower-level data that is often not normalized—and *indices*, which aggregate indicators and are normalized on a scale between zero and one.

There are four datasets: *Coder-Level*, with raw data and 273 indicators intended for the experts; *CD Country-Date*, with day-to-day granularity, 531 indicators, and 95 aggregated indices; *CY Country-Year*, with year-level granularity, 5 indices, 93 sub-indices, and 179 indicators; and lastly, *CY Full + Others*, with year-level granularity, 531 indicators, 251 indices, and 62 external indicators.

I must use the *CD Country-Date* dataset (version 16, released in March 2026) since, to keep track of Italian governments, year-level granularity is not enough. This is evident from Figure 3: half of Italy's governments lasted less than a year.

The dataset covers Italy (ID 82) between 1861 and 2025, France (ID 76) between 1789 and 2025, and the U.S. (ID 20) between 1789 and 2025.

Useful identifiers include: *country_id*, unique for each country; *country_text_id*, ISO 3166-1 alpha-2 country codes (e.g., *IT*, *FR*); *historical_date*, in YYYY-MM-DD format, used for election-date-specific variables (all other variables use January 1st); and *gapstart* and *gapend*, representing time periods where no data is available.

Democracy Indices are the highest-level indices, each normalized to a scale between zero and one unless otherwise specified: *v2x_polyarchy*, the *electoral* democracy index (aggregates *v2x_elecoff*, *v2xel_frefair*, *v2x_frassoc_thick*, *v2x_suffr*, *v2x_freexp_altinf*); *v2x_libdem*, the *liberal* democracy index (aggregates *v2x_polyarchy*, *v2x_liberal*); *v2x_partipdem*, the *participatory* democracy index (aggregates *v2x_polyarchy*, *v2x_partip*); *v2x_delibdem*, the *deliberative* democracy index focusing on process, respectful dialogue, and the common good (aggregates *v2x_polyarchy*, *v2xd1_delib*); and *v2x_egaldem*, the *egalitarian* democracy index considering social status, inequalities, and resource distribution (aggregates *v2x_polyarchy*, *v2x_egal*).

Components of Democracy Indices consist of second-level indices and subcomponents: *v2x_freexp_altinf*, freedom of expression and alternative information (aggregates *v2mecenefm*, *v2meharjrn*, *v2meslfcen*, *v2xcl_disc*, *v2clacfree*, *v2mebias*, *v2mecrit*, *v2merange*); *v2x_frassoc_thick*, freedom of association for parties and civil society organizations

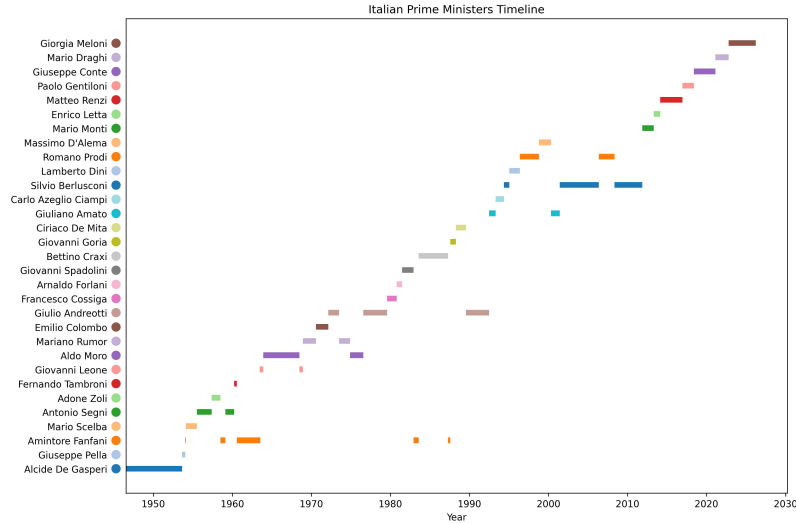


Fig. 3. Italian Prime Ministers for each Italian Republic, from the first (1946) to the latest (2025).

(aggregates *v2psparban*, *v2psbars*, *v2psoppaut*, *v2elmulpar*, *v2cseeorgs*, *v2csreprss*, *v2x_elecreg*); *v2x_suffr*, the percentage of adult suffrage (aggregates *v2elsuffrage*); *v2xel_frefair*, the quality of free and fair elections (aggregates *v2elembaut*, *v2elembcap*, *v2elrgstry*, *v2elvotbuy*, *v2elirreg*, *v2elintim*, *v2elpeace*, *v2elfrfair*, *v2x_elecreg*); *v2xcl_rol*, rule of law, transparency, and fair enforcement (aggregates *v2clrspct*, *v2cltrnslw*, *v2cltort*, *v2clkill*, *v2clrelig*, *v2clfmove*, *v2xcl_dmove*, *v2xcl_slave*, *v2xcl_acjst*, *v2xcl_prpty*); *v2x_jucon*, executive law compliance and judiciary independence (aggregates *v2exrescon*, *v2jucomp*, *v2juhccomp*, *v2juhcind*, *v2juncind*); *v2xlg_legcon*, executive external scrutiny (aggregates *v2lgqstexp*, *v2lgotovst*, *v2lginvtst*, *v2lgoppart*); *v2x_partip*, individual participation (aggregates *v2x_cspart*, *v2xdd_dd*, *v2xel_locelec*, *v2xel_regelec*); *v2x_cspart*, participation in associations like labor unions or NGOs (aggregates *v2pscnslnl*, *v2cscnsult*, *v2csptrcpt*, *v2csgender*); *v2xdd_dd*, the direct vote index for referendums and ballots (aggregates *v2ddlexci* through *v2ddthreci*); *v2xel_regelec* and *v2xel_locelec*, regional and local government elections; *v2xeg_eqprotec*, equal rights across social groups; *v2xeg_eqaccess*, equality of accessing power; and *v2xeg_eqdr*, equality of resource distribution (aggregates *v2dlencmps*, *v2dlunivl*, *v2peedueq*, *v2pehealth*).

7 MODEL FINE-TUNING

Classification tasks can be performed, broadly speaking, in two ways: the first leverages common machine learning and statistical tools, such as logistic regression; the second approach uses Transformer models (commonly referred to today as "AI"). A traditional approach might involve using TF-IDF and logistic regression with the Scikit-Learn library (example code at 1).

Listing 1. Scikit-Learn Logistic Regression for classification task

```
1 from sklearn.feature_extraction.text import TfidfVectorizer
2 from sklearn.linear_model import LogisticRegression
3 from sklearn.pipeline import Pipeline
4
5 X_train, X_test, y_train, y_test = train_test_split(df['text'], df['label'], test_size=0.2)
```

```

521 6
522 7 text_clf = Pipeline([
523 8     ('tfidf', TfidfVectorizer(stop_words='english', ngram_range=(1, 2))),
524 9     ('clf', LogisticRegression())
525 10 ])

```

On the other hand, Transformers are far superior in terms of accuracy for textual analysis, although they are much more resource-demanding. An example would be using Hugging Face's transformers library to fine-tune a pre-trained model (example code at 2).

Listing 2. Hugging Face Trainer for fine-tuning

```

534 1 from transformers import AutoModelForSequenceClassification, AutoTokenizer, Trainer, TrainingArguments
535 2
536 3 trainer = Trainer(
537 4     model=AutoModelForSequenceClassification.from_pretrained(model_name, num_labels=2),
538 5     args=training_args,
539 6     train_dataset=tokenized_datasets["train"],
540 7     eval_dataset=tokenized_datasets["test"],
541 8 )
542 9
543 10 trainer.train()

```

Regarding the literature on the subject, Card et al. [7] fine-tuned a pre-trained RoBERTa model on congressional speeches in a self-supervised fashion and then further fine-tuned it as a classifier using annotated examples. They achieved 65% accuracy in classifying Congressional Record entries as being pro-immigration, anti-immigration, or neutral.

Albladi et al. [1] surveyed the field of NLP sentiment analysis and found that RoBERTa consistently outperforms other methods in multi-classification tasks. Their results revealed a clear correlation between dataset size and improved model performance, with larger datasets significantly increasing both accuracy and F1 scores across all models. For example, GPT-3.5 reached accuracies of 0.9694, 0.9784, 0.9797, and 0.9875 for various dataset sizes (e.g., 500, 1000, and 2500). However, LLaMA-2 13B emerged as the most balanced option for NLP tasks, striking an optimal balance between computational efficiency, performance, and adaptability. This model can be fine-tuned on freely available GPUs, such as the T4 on Google Colab, using techniques like QLoRA, which makes it an attractive choice for researchers and developers.

Li et al. [23] found that LS-LLaMA substantially outperforms LLMs ten times its size in text classification tasks and demonstrates consistent improvements compared to robust baselines like BERT-Large and RoBERTa-Large.

Considering all of this, it would be interesting to fine-tune a Mistral 7B model, which is claimed to perform better than LLaMA 7B and LLaMA 13B across all benchmarks (Figure 4).

8 FEASIBILITY

The project can be completed within the three-year duration of the PhD program. One year is expected to be dedicated to completing the dataset, leaving sufficient time to conduct the ensuing textual analyses.

Furthermore, the majority of the work required for the Italian corpora will be completed in the coming months as part of my Master's thesis project.

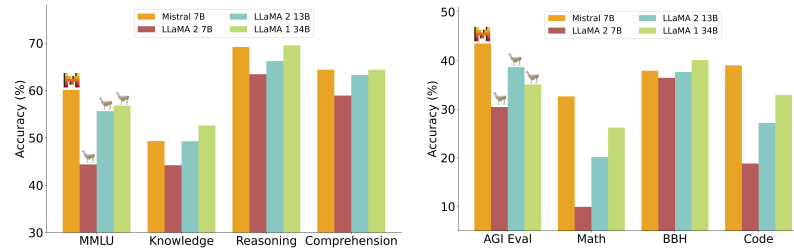


Fig. 4. Mistral 7B consistently outperforms LLaMA on all tasks. Source: Mistral.

This work will tie neatly to my career progression since I changed my specialization from cybersecurity to a more data analysis-oriented path. I also discovered an interest in AI, and specifically natural language processing through transformer models, which this PhD will allow me to explore.

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